CHAPTER 6

Oxidation and Reduction

Oxidation Number (O.N.)

The oxidation number of an atom in a molecule or ion is a number which indicates qualitatively its state of oxidation. The number is determined by applying the following rules:

1. For elements in the free form, the oxidation number is 0.
2. For monatomic ions, the oxidation number is the charge on the ion.
3. For combined oxygen, the oxidation number is -2, except in peroxides such as H_2O_2 and Na_2O_2 where it is -1.
4. For combined hydrogen, the oxidation number is +1, except in metal hydrides, where it is -1.
5. For a molecule or ion, the sum of the oxidation numbers of the constituent atoms is zero for a molecule or is the charge for an ion.

Examples

1. What is the oxidation number of chlorine in potassium perchlorate ($KClO_4$)?

	K	Cl	O_4		
O.N.	(+1)	?	(-2)		
Since	(+1)	+ (?) +	4(-2)	=	0
Then			O.N.(Cl)	=	+7

2. Determine the oxidation number of sulfur in the hydrogensulfite ion (HSO_3^-)

	H	S	O_3^-		
O.N.	(+1)	?	(-2)		
Since	(+1)	+ (?) +	3(-2)	=	-1
Then			O.N.(S)	=	+4

Changes in Oxidation Number

Oxidation involves an increase in oxidation number, while reduction involves a decrease in oxidation number.

In an oxidation-reduction reaction oxidation and reduction occur simultaneously. One element must undergo an increase in oxidation number, while another undergoes a decrease in oxidation number.

If none of the oxidation numbers change, then the reaction is **not** an oxidation-reduction reaction.

To identify which elements have changed oxidation number the following rules are applied:

1. Ignore the coefficients of substances in the equation.
2. Identify the oxidation numbers of each of the elements.
3. Determine which elements have changed their oxidation number.

Example

1. Determine which elements have undergone oxidation and reduction in the reaction represented by the equation:

$$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$$

NH_3	+	O_2	$\rightarrow$	NO	+	H_2O
(-3)(3 x (+1))		(0)		(+2)(-2)		(2 x (+1))(-2)

N (-3) $\rightarrow$ (+2)

O (0) $\rightarrow$ (-2)

Nitrogen oxidised, (-3) $\rightarrow$ (+2)

Oxygen reduced, (0) $\rightarrow$ (-2)

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1. Determine the oxidation number of the bolded element in each of the following:

(a)	SO_2	(b)	H_2S
(c)	H_2SO_4	(d)	$Na_2S_2O_3$
(e)	SF_6	(f)	P_2O_5
(g)	PH_3	(h)	$[Cu(NH_3)_4]^{2+}$
(i)	H_3PO_4	(j)	$Mg_2P_2O_7$
(k)	CH_4	(l)	CO_2
(m)	CH_3OH	(n)	$HCHO$
(o)	$HCOOH$	(p)	NO_2
(q)	N_2O	(r)	NH_4Cl
(s)	$NaNO_3$	(t)	N_2H_4
(u)	$SnCl_4$	(v)	SnO
(w)	Cu_2O	(x)	CuS
(y)	$FeCl_2$	(z)	Fe_2O_3

2. For each of the following reactions determine whether any of the elements have undergone a change in oxidation number and note whether they have been oxidised or reduced.

(a) $2Mg + O_2 \rightarrow 2MgO$

(b) $2Na + 2H_2O \rightarrow 2NaOH + H_2$

(c) $Cl_2 + 2I^- \rightarrow I_2 + 2Cl^-$

(d) $3H_2S + 2HNO_3 \rightarrow 3S + 2NO + 4H_2O$

(e) $2MnO_4^- + 5H_2S + 6H^+ \rightarrow 2Mn^{2+} + 5S + 8H_2O$

(f) $2SnCl_2 + 4HCl + O_2 \rightarrow 2SnCl_4 + 2H_2O$

(g) $H_2SO_4 + 2HBr \rightarrow SO_2 + Br_2 + 2H_2O$

(h) $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

(i) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

(j) $NH_4NO_2 \rightarrow N_2 + 2H_2O$

(k) $Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4$

(l) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$

(m) $HCl + NaOH \rightarrow NaCl + H_2O$

(n) $H_2S + CuCl_2 \rightarrow CuS + 2HCl$

(o) $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$

(p) $CaO + H_2O \rightarrow Ca(OH)_2$

(q) $Cl_2 + 2KBr \rightarrow Br_2 + 2KCl$

(r) $2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$

Balancing Redox Equations

Redox equations are often complex and a special method, the half-equation method, is used to balance them.

Balancing Half Equations

To balance a half-equation the following procedure is used:

1. Set out the reactant and product, together with the respective oxidation numbers, in equation form.

2. Balance the number of atoms of the element which is oxidised or reduced.

3. Calculate the change in oxidation number of the element from reactant to product. If there is an increase in oxidation number, add an equal number of electrons to the right- hand side of the equation. If there is a decrease in oxidation number add the electrons to the left-hand side of the equation. If more than one atom is oxidised or reduced, the number of electrons added to the half equation equals the number of atoms of the element multiplied by the change in oxidation number.

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1.

(a)	+4	(b)	-2
(c)	+6	(d)	+2
(e)	+6	(f)	+5
(g)	-3	(h)	+2
(i)	+5	(j)	+5
(k)	-4	(l)	+4
(m)	-2	(n)	0
(o)	+2	(p)	+4
(q)	+1	(r)	-3
(s)	+1	(t)	-2
(u)	+4	(v)	+2
(w)	+1	(x)	+2
(y)	+2	(z)	+3

2.

	Element Oxidised	Element Reduced
(a)	Mg(0 $\rightarrow$ +2)	O(0 $\rightarrow$ -2)
(b)	Na(0 $\rightarrow$ 1)	H(+1 $\rightarrow$ 0)
(c)	I(-1 $\rightarrow$ 0)	Cl(0 $\rightarrow$ -1)
(d)	S(-2 $\rightarrow$ 0)	N(+5 $\rightarrow$ +2)
(e)	S(-2 $\rightarrow$ 0)	Mn(+7 $\rightarrow$ +2)
(f)	Sn(+2 $\rightarrow$ +4)	O(0 $\rightarrow$ -2)
(g)	Br(-1 $\rightarrow$ 0)	S(+6 $\rightarrow$ +4)
(h)	Fe(+2 $\rightarrow$ +3)	Cr(+6 $\rightarrow$ +3)
(i)	None (not a redox reaction)	
(j)	N(-3 $\rightarrow$ 0)	N(+3 $\rightarrow$ 0)
(k)	None (this is a precipitation reaction)	
(l)	None (This is an acid/carbonate equation)	
(m)	None (This is an acid/base neutralisation equation)	
(n)	None	
(o)	None	
(p)	None	
(q)	Br(-1 $\rightarrow$ 0)	Cl(0 $\rightarrow$ -1)
(r)	Br(-1 $\rightarrow$ 0)	S(+6 $\rightarrow$ +4)

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1. $Mg \rightarrow Mg^{2+} + 2e^-$
2. $S + 2e^- \rightarrow S^{2-}$
3. $2Br^- \rightarrow Br_2 + 2e^-$
4. $Cl_2 + 2H_2O \rightarrow 2ClO^- + 4H^+ + 2e^-$
5. $6H_2O + Br_2 \rightarrow 2BrO_3^- + 12H^+ + 10e^-$
6. $NO_3^- + 4H^+ + 3e^- \rightarrow NO + H_2O$